Comparison of the Microsoft Kinect V2 and FARO Focus 120 for 3D Plant Phenotyping of Cassava (*Manihot esculenta*)

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Introduction and Rationale

In recent years, remote sensing applications have been increasingly explored as a means of determining plant phenotype. Current literature suggests that 3D phenotyping can be successful, with studies showing significant correlation to biomass, plant height, and other phenotypic traits. However, two issues limit the widespread use of many 3D plant phenotyping platforms:

1) Complexity of the data analysis process
2) Substantial platform acquisition costs

We attempted to overcome the cost limitation by using an alternative sensor.

Originally released as a video game accessory to allow hands-free gaming for Microsoft's X-Box consoles, the **Microsoft Kinect V2** sensor is an RGB-D camera that allows collection of high-resolution 3D data using an infrared (IR) time-of-flight sensor, and has shown promise as a scientific tool, especially due to its low cost and high portability.

Objectives and Hypothesis

The objective of this experiment is to determine usability of the Microsoft Kinect V2 as a 3D modelling tool and assess its usefulness as a substitute for TLS in plant breeding. This was achieved through a direct comparison of the Kinect's pointcloud with that of a FARO Focus 120, an industry-standard terrestrial laser scanner (Table 1).

Table 1. Key Features of the Microsoft Kinect V2 and the Faro Focus 120.

Feature	Microsoft Kinect V2	Faro Focus 120
Type	RGB-D	Terrestrial Laser Scanner
Wavelength (nm)	860	905
FOV (H X V)	70° X 60°	360° X 300°
Range (m)	~0.8 – 4.5	~0.6 – 120
Depth Pixel Array (H X V)	512 X 424	
Color Camera	1920 X 1080	70 megapixel
Frames Per Second	30	

We expected that the FARO scanner would produce a much clearer pointcloud, but that the Kinect would capture an accurate representation of the plant's biomass in a fraction of the time.

Methodology

Images of six cassava plants of three varieties (7 months old) were captured in a greenhouse at Texas A&M University. For each plant, four scans were taken with the FARO and the Kinect V2 at 0°, 90°, 180°, 270°, using a turntable to facilitate rotation (Figure 1). The cameras were placed 90° to one another surrounding the plant at a distance of 1.3 m and a height of 0.97 m.

Figure 1. Microsoft Kinect V2 (left) and the Faro Focus 120 (right).

Processing was done in CloudCompare (CC) software:
- Clipping
- Noise Removal
- Registration
- Pointcloud Merging

Basic Analysis was also done in CC:
- Cloud-to-Cloud distance between the FARO and Kinect models
- Assessments of models with pot and without
- Welch T-Test to compare the difference
- Point density (based on number of neighbors inside a sphere of 0.01.
- A Welch T-Test for variances

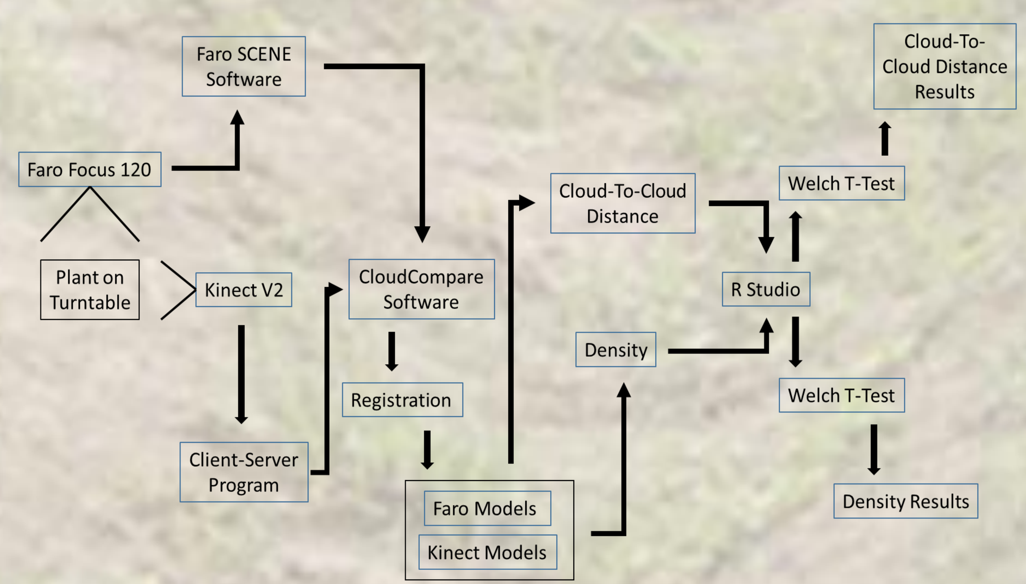

Figure 2. Outline of the data collection and analysis workflow.

Results

Cloud-to-Cloud Distance

The average distance between the pointclouds, based on the means of all plants, was 0.0093 m (SD 0.0013 m) (Figure 3). All mean pointcloud distances were less than 1 cm except for plant 3, which was 0.011 m. The spatial distribution of the variability in distance between the pointclouds for all plants was evenly distributed with two exceptions (Figure 4):

1) Points on the positive tail of the distribution (red) seem to be grouped largely within the vegetation.
2) The pots seem to produce concentrations of points which have above-average distances (0.15 and 0.04 m).

Cloud-to-Cloud distance analysis was again performed after removing non-vegetation points (Figure 5). The average distance between the pointclouds, based on the means of all plants, was 0.0090 m (SD 0.0019 m). The Welsh T-Test (Table 2) indicates that the removal of the pot made a statistically significant impact for all plants except number 6. However, the change in the mean distance was both positive and negative depending on the plant, and was less than 0.001 m except for plant 2, where the mean increased by 0.002 m.

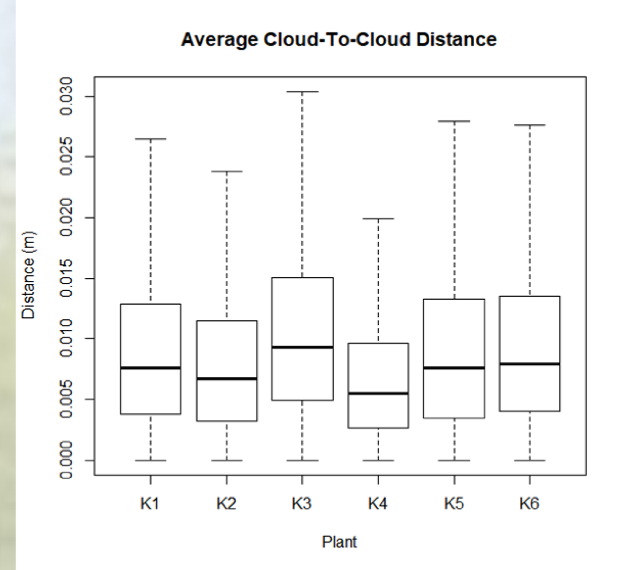

Figure 3. Distance from Faro model to the Kinect model.

Table 2. Distance from the Faro model to the Kinect model with the pot removed.

Plant	t-value	df	p-value	Change in Mean
1	6.5295	74476	6.641e-11	0.000335
2	18.251	11699	2.2e-16	0.002025
3	-2.6336	41714	0.008452	-0.000245
4	3.5472	58792	0.0003896	0.000204
5	-9.9175	61167	2.2e-16	-0.000628
6	1.3178	43279	0.1876	0.000094

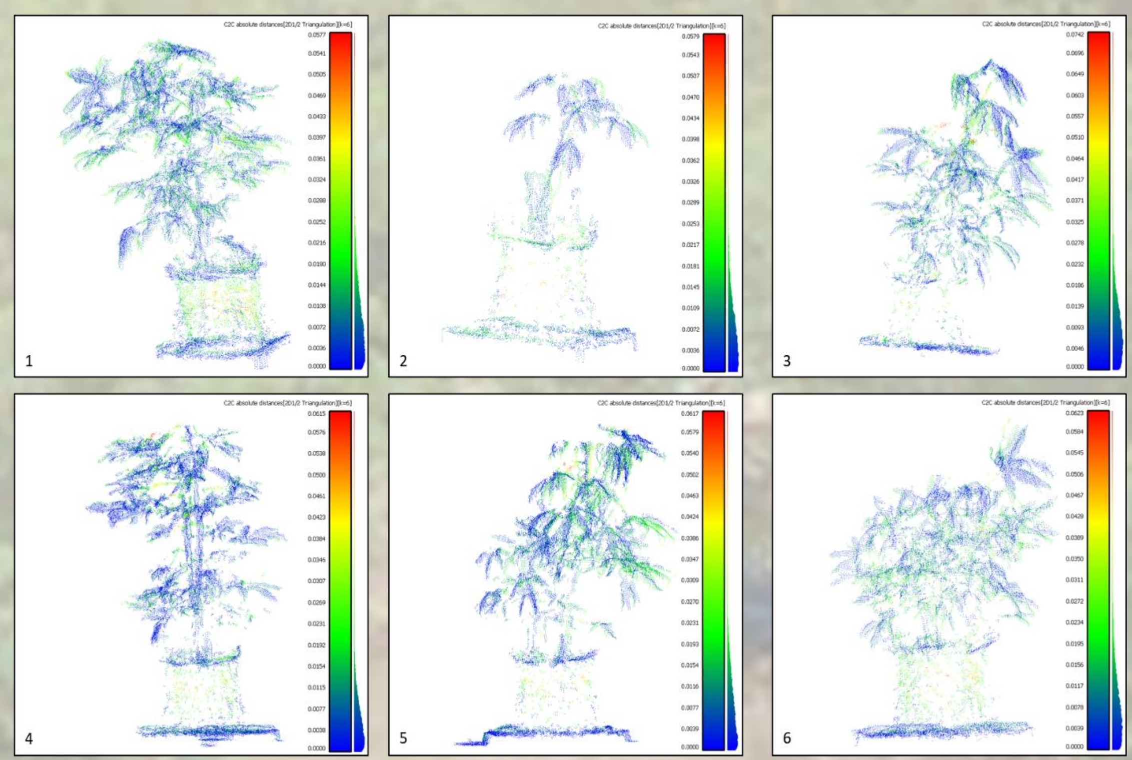

Figure 4. The Cloud-to-Cloud distance for each Kinect model visualized in CloudCompare.

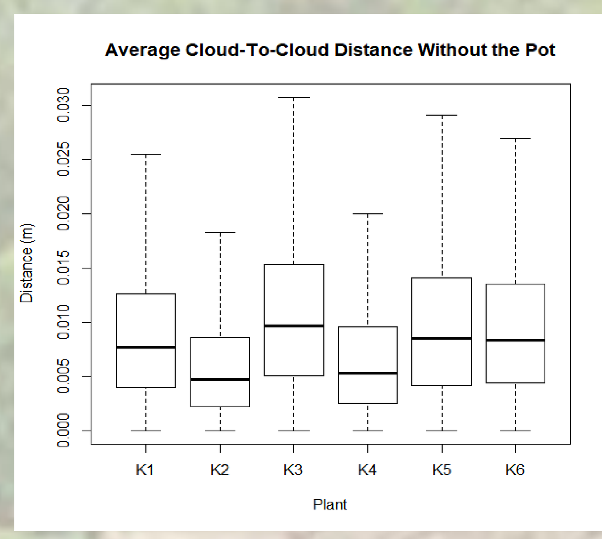

Figure 5. Distance from the Faro model to the Kinect model with the pot removed.

Point Density

Point density analysis for each plant showed that the FARO scanner had a significantly greater point density than the Kinect (Table 3). The average point density for the Kinect models was 14.66 neighbors versus 74.43 neighbors for the Faro models within a radius of 0.01 m. This resulted in an average difference in the mean point density of 59.8 (Figure 6).

Table 3. Welsh T-Test comparing the pointcloud density between the Faro and Kinect models.

Plant	t-value	df	p-value	Difference in Mean
1	797.85	248440	2.2e-16	55.11916
2	505.81	53850	2.2e-16	59.52009
3	573.57	164590	2.2e-16	74.3083
4	673	229100	2.2e-16	61.60727
5	694.39	206030	2.2e-16	56.74021
6	665.68	151550	2.2e-16	51.8106

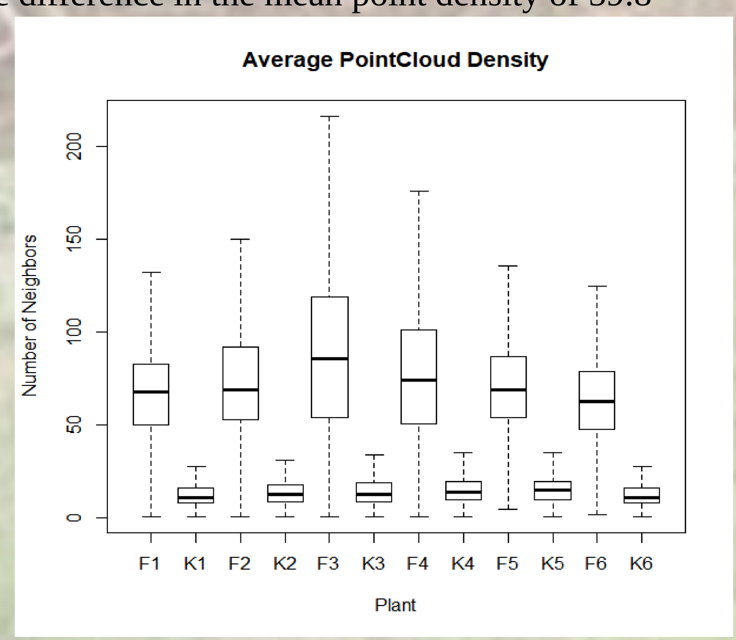

Figure 6. The average pointcloud density for the Faro and Kinect models.

Discussion

The average Cloud-to-Cloud distance was ~1 cm, in agreement with the error found by Lachat et al., 2015. Mean error might be improved if points with ~ 3 cm were removed. Many of these points were concentrated around single areas in the vegetation (suggesting slight leaf movement). Nevertheless, an error of ~ 1 cm should have little effect on many high-throughput phenotyping tasks, and more accurate measures by hand are unlikely.

The FARO models are approximately 5 times denser than that of the Kinect (14.66 neighbors within a 0.01 m radius for Kinect and 74.43 for FARO). It is was clear that the black plastic pots were not captured with much spatial accuracy or point density in the Kinect models. However, major plant parts (leaves, stems, and main stalk) are represented well, despite the lower density. Precise estimations of fine physiological features might produce larger errors as compared to the FARO, and point density from Kinect scans will decrease with sensor distance from the subject. This could have a major impact on data quality for very large plants or those captured at greater distances.

Figure 7. A side-by-side comparison of Plant 4; Kinect on the left, Faro on the right.

Conclusion

Despite the lower resolution, the Kinect can be a cost-effective alternative for applications that don't require sub-centimeter resolutions (e.g. height or volume approximations) (Table 4). These sensors also benefit from being highly portable, and could be easily attached to many different field cart designs.

Table 4. Strengths and weaknesses of the Kinect and FARO.

Feature	Microsoft Kinect V2	Faro Focus 120
Bright Sunlight		✓
Windy Conditions	✓	
Time Efficient	✓	
Extended Range		✓
Low-cost	✓	
Commercial Product		✓
High Pointcloud Density	✓	✓

Identification of aboveground material may be key to determining superior cultivars of cassava in the future. High-throughput phenotyping could be an invaluable tool in determining foliar traits that correspond to early storage root bulking, drought and leaf retention rates, etc., thus improving the efficiency of the crop. The low cost and easy transportation of the Kinect could open the door for breeders to implement this technology even on a tight budget.

The next step will be to create a field-worthy platform to test the Kinect sensors outside (Figures 8 and 9).

Figure 8. The "Scorpion" phenotyping platform.

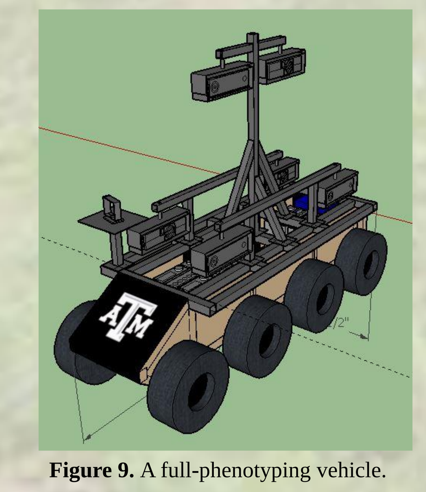

Figure 9. A full-phenotyping vehicle.

These upgrades will allow the Kinect to be used alongside traditional laser scanners in a field setting, where biomass assessments and sensor responses to different colors and/or materials can be more thoroughly evaluated.

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